Movement Simulation Methods and Results

Simulation Methods

1. Generating Movement Trajectories

We simulated continuous-time animal movement trajectories using Langevin diffusion models to evaluate the recovery of resource selection coefficients under varying sampling and discretization conditions. Simulations were performed under two movement paradigms: an underdamped model (which incorporates momentum via a friction parameter $\gamma = 0.5$) and an overdamped model (representing pure 2D Brownian motion without directional persistence).

For each scenario, we generated 100 independent tracks consisting of 100,000 true locations at a fine temporal resolution ($\Delta t = 0.01$). The baseline diffusion (or speed) parameter was set to $\sigma = 2.5$ for the "High-Gradient / Fast-Diffusion" scenario and $\sigma = 1$ for the "Low-Gradient / Slow-Diffusion" scenario.

2. Spatial Environment & Barriers

The true habitat landscape was generated on a fine spatial grid (`langRes = 1`) bounded by a spatial scale of 200 units. To construct a realistic, heterogeneous environment, three continuous spatial covariates were generated with random spatial ranges to represent autocorrelated resource fields. A quadratic distance-to-center penalty (`d2c`) was included to prevent tracks from drifting infinitely away from the origin.

To evaluate the effects of absolute movement boundaries, a complex coastal barrier was simulated, containing a mainland boundary and randomly scattered islands. This generated a discrete land mask and a continuous distance-to-coastline covariate. True resource selection coefficients driving the movement were set to $\beta = (-4, 6, 5, -0.1, -0.1)$ for the High-Gradient / Fast-Diffusion scenario, and $\beta = (-1, 2, 0, -0.1, -0.1)$ for the Low-Gradient / Slow-Diffusion scenario (representing the three resource covariates, `d2c`, and `d2coast`, respectively).

3. Observation Process and Measurement Error

To mimic real-world telemetry arrays, the true continuous paths were subsampled to introduce significant missing data and spatial uncertainty. We applied missing data proportions of 0.99, 0.995, and 0.998, which translated to effective sampling intervals (dt) of 1, 2, and 5 time units.

Telemetry measurement error was applied as a bivariate normal distribution. The semi-major axis standard deviation (σ_{err}) was varied across scenarios (0.5, 2.5, 5, and 10), with the semi-minor axis defined as half the semi-major axis, and error ellipse orientations drawn from a uniform distribution (0–180 degrees).

4. Model Fitting & Spatial Discretization Pipeline

Subsampled tracks were fit using two models: the true continuous-time Langevin model, and a discretized Continuous-Time Markov Chain (CTMC) model utilizing the `walk` package. To investigate discretization artifacts, the CTMC model was fit at varying spatial resolutions (`walkRes` $\in \{1, 2, 5, 8\}$).

Because aggregating rasters can inherently destroy fine-scale topologies (such as narrow land bridges or small islands), a rigorous projection pipeline was used to evaluate utilization distributions (UDs). The coarse

`{walk}` model UD was zero-padded using `terra::extend` to prevent bounding-box truncation artifacts. The padded raster was then projected back to the fine `langRes = 1` grid via bilinear interpolation to reconstruct smooth continuous probability gradients. Finally, the exact high-resolution coastline mask was applied post-hoc to perfectly preserve physical barriers, and the resulting UD was renormalized to sum to 1.

5. Evaluation Metrics

Model performance was quantified by comparing the Estimated UD to the True UD across the valid fine-scale grid. We evaluated three metrics: Bhattacharyya's Affinity (BA), which measures the colocation and pattern similarity between the true and estimated probability distributions (0 = no similarity, 1 = complete similarity); Schoener's D, which measures the absolute probability intersection (0 = no intersection, 1 = complete intersection); and the Root Mean Square Error of the log-transformed UDs (log RMSE), which captures deviations in magnitude and intensity (< 0.5 = tight fit, $0.5 - 1.5$ = good fit, > 2 = poor fit).

Results

Table 1: Effect of walkRes on Bhattacharyya's Affinity (BA) ($\sigma_{err} = 2.5$)

			dt = 1		dt = 2		dt = 5	
Scenario	Model	walkRes	walk	lang	walk	lang	walk	lang
Low-Gradient / Slow-Diffusion	Underdamped	1	0.876	0.945	0.886	0.944	0.911	0.940
		2	0.893	0.945	0.897	0.944	0.911	0.940
		5	0.918	0.945	0.909	0.944	0.911	0.937
		8	0.930	0.946	0.914	0.944	0.913	0.941
	Overdamped	1	0.811	0.885	0.848	0.924	0.811	0.904
		2	0.829	0.885	0.853	0.924	0.811	0.904
		5	0.863	0.883	0.869	0.924	0.822	0.905
		8	0.871	0.886	0.870	0.922	0.836	0.908
High-Gradient / Fast-Diffusion	Underdamped	1	0.961	0.712	0.973	0.678	0.966	0.691
		2	0.949	0.707	0.959	0.678	0.951	0.691
		5	0.929	0.712	0.933	0.683	0.926	0.689
		8	0.905	0.712	0.907	0.687	0.902	0.691
	Overdamped	1	0.958	0.705	0.960	0.713	0.952	0.715
		2	0.943	0.703	0.947	0.711	0.940	0.715
		5	0.914	0.703	0.916	0.711	0.915	0.714
		8	0.894	0.706	0.895	0.712	0.896	0.714

Table 2: Effect of walkRes on Schoener's D ($\sigma_{err} = 2.5$)

			dt = 1		dt = 2		dt = 5	
Scenario	Model	walkRes	walk	lang	walk	lang	walk	lang
Low-Gradient / Slow-Diffusion	Underdamped	1	0.654	0.761	0.669	0.755	0.709	0.746
		2	0.678	0.761	0.680	0.755	0.708	0.746
		5	0.708	0.761	0.686	0.755	0.696	0.743
		8	0.726	0.763	0.691	0.753	0.699	0.749
	Overdamped	1	0.566	0.672	0.606	0.735	0.565	0.709
		2	0.586	0.672	0.610	0.735	0.565	0.709
		5	0.617	0.670	0.628	0.733	0.571	0.711
		8	0.623	0.674	0.631	0.730	0.585	0.713
High-Gradient / Fast-Diffusion	Underdamped	1	0.822	0.411	0.840	0.382	0.820	0.395
		2	0.763	0.408	0.786	0.382	0.769	0.395
		5	0.713	0.412	0.718	0.386	0.707	0.393
		8	0.663	0.413	0.665	0.390	0.658	0.395
	Overdamped	1	0.801	0.407	0.807	0.410	0.784	0.414
		2	0.751	0.406	0.762	0.408	0.744	0.414
		5	0.683	0.405	0.691	0.408	0.690	0.413
		8	0.643	0.408	0.648	0.409	0.651	0.413

Table 3: Effect of walkRes on log(RMSE) ($\sigma_{err} = 2.5$)

			dt = 1		dt = 2		dt = 5	
Scenario	Model	walkRes	walk	lang	walk	lang	walk	lang
Low-Gradient / Slow-Diffusion	Underdamped	1	1.588	1.014	1.515	0.981	1.170	0.935
		2	1.452	1.014	1.469	0.981	1.168	0.935
		5	1.215	1.008	1.332	0.973	1.178	0.964
		8	1.088	0.992	1.266	0.975	1.290	0.959
	Overdamped	1	2.096	1.481	1.692	1.021	2.175	1.196
		2	1.915	1.481	1.656	1.020	2.180	1.196
		5	1.642	1.498	1.545	1.034	2.032	1.193
		8	1.563	1.509	1.488	1.047	1.880	1.181
High-Gradient / Fast-Diffusion	Underdamped	1	0.847	2.522	0.785	2.806	0.860	2.740
		2	0.953	2.563	0.877	2.806	0.939	2.740
		5	1.048	2.549	1.011	2.775	1.085	2.770
		8	1.270	2.556	1.195	2.738	1.248	2.761
	Overdamped	1	0.843	2.442	0.849	2.408	0.989	2.350
		2	0.958	2.452	0.942	2.408	1.052	2.354
		5	1.139	2.468	1.117	2.402	1.192	2.356
		8	1.235	2.427	1.272	2.394	1.298	2.343

Table 4: Effect of Measurement Error on Bhattacharyya's Affinity (BA) (walkRes = 2)

			dt = 1		dt = 2		dt = 5	
Scenario	Model	σ_{err}	walk	lang	walk	lang	walk	lang
Low-Gradient / Slow-Diffusion	Underdamped	0.5	0.889	0.945	0.912	0.950	0.914	0.939
		2.5	0.893	0.945	0.897	0.944	0.911	0.940
		5	0.889	0.939	0.885	0.928	0.904	0.942
		10	0.875	0.919	0.869	0.924	0.881	0.939
	Overdamped	0.5	0.881	0.890	0.860	0.868	0.842	0.879
		2.5	0.829	0.885	0.853	0.924	0.811	0.904
		5	0.814	0.894	0.795	0.895	0.835	0.932
		10	0.752	0.870	0.767	0.888	0.764	0.913
High-Gradient / Fast-Diffusion	Underdamped	0.5	0.950	0.842	0.957	0.795	0.948	0.733
		2.5	0.949	0.707	0.959	0.678	0.951	0.691
		5	0.956	0.664	0.948	0.675	0.944	0.695
		10	0.937	0.620	0.945	0.650	0.932	0.684
	Overdamped	0.5	0.951	0.792	0.947	0.781	0.932	0.729
		2.5	0.943	0.703	0.947	0.711	0.940	0.715
		5	0.942	0.686	0.945	0.690	0.944	0.702
		10	0.925	0.664	0.934	0.691	0.920	0.678

Table 5: Effect of Measurement Error on Schoener's D (walkRes = 2)

			dt = 1		dt = 2		dt = 5	
Scenario	Model	σ_{err}	walk	lang	walk	lang	walk	lang
Low-Gradient / Slow-Diffusion	Underdamped	0.5	0.675	0.768	0.711	0.771	0.709	0.744
		2.5	0.678	0.761	0.680	0.755	0.708	0.746
		5	0.679	0.741	0.659	0.722	0.691	0.749
		10	0.653	0.709	0.648	0.716	0.653	0.741
	Overdamped	0.5	0.649	0.672	0.630	0.645	0.588	0.649
		2.5	0.586	0.672	0.610	0.735	0.565	0.709
		5	0.568	0.675	0.534	0.681	0.591	0.737
		10	0.488	0.639	0.515	0.670	0.502	0.699
High-Gradient / Fast-Diffusion	Underdamped	0.5	0.769	0.566	0.778	0.503	0.765	0.438
		2.5	0.763	0.408	0.786	0.382	0.769	0.395
		5	0.778	0.366	0.767	0.378	0.756	0.396
		10	0.747	0.331	0.752	0.358	0.737	0.386
	Overdamped	0.5	0.765	0.499	0.758	0.488	0.739	0.431
		2.5	0.751	0.406	0.762	0.408	0.744	0.414
		5	0.747	0.388	0.754	0.389	0.751	0.404
		10	0.720	0.369	0.732	0.393	0.717	0.381

Table 6: Effect of Measurement Error on log(RMSE) (walkRes = 2)

			dt = 1		dt = 2		dt = 5	
Scenario	Model	σ_{err}	walk	lang	walk	lang	walk	lang
Low-Gradient / Slow-Diffusion	Underdamped	0.5	1.440	0.964	1.269	0.945	1.355	1.104
		2.5	1.452	1.014	1.469	0.981	1.168	0.935
		5	1.647	1.105	1.742	1.111	1.292	0.949
		10	1.959	1.331	1.921	1.272	1.640	0.979
	Overdamped	0.5	1.465	1.476	1.625	1.599	1.638	1.373
		2.5	1.915	1.481	1.656	1.020	2.180	1.196
		5	2.359	1.559	2.699	1.449	1.964	1.001
		10	3.900	2.301	3.058	1.592	3.202	1.282
High-Gradient / Fast-Diffusion	Underdamped	0.5	0.911	1.894	0.913	2.158	0.914	2.416
		2.5	0.953	2.563	0.877	2.806	0.939	2.740
		5	1.016	2.910	0.977	2.849	1.106	2.759
		10	1.268	3.448	1.141	3.202	1.195	2.939
	Overdamped	0.5	0.911	2.152	0.976	2.126	1.041	2.342
		2.5	0.958	2.452	0.942	2.408	1.052	2.354
		5	1.135	2.681	1.100	2.564	1.135	2.641
		10	1.448	3.088	1.275	2.850	1.322	2.820